THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 5: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two circles of radii 13 cm and 20 cm intersect in a common chord of length 24 cm and their centres lie on opposite sides of it. The distance between the centres (in cm) is: 1

(a) 20 **(b)** 23 **(c)** 22 **(d)** 21

2. $ABCD$ is cyclic, $\angle ABD = 42^\circ$ and $\angle DBC = 38^\circ$. Then $\angle ABC =$ 1

(a) 90° **(b)** 80° **(c)** 100° **(d)** 70°

3. The exterior angle at D of cyclic quadrilateral $ABCD$ (side CD produced) equals the interior angle at B, which is 85°. The interior angle $\angle ADC$ is: 1

(a) 95° **(b)** 115° **(c)** 85° **(d)** 105°

4. AB is a diameter, C and D are on the circle (same side of AB) and $ABCD$ is cyclic with $\angle DAB = 57^\circ$. Then $\angle BCD =$ 1

(a) 123° **(b)** 133° **(c)** 143° **(d)** 113°

5. A chord AB of length 30 cm lies 20 cm from the centre O. The area of triangle OAB (cm^2) is: 1

(a) 299 **(b)** 301 **(c)** 300 **(d)** 302

6. If a parallelogram is inscribed in a circle, it must be a: 1

(a) rhombus **(b)** trapezium **(c)** square only **(d)** rectangle

7. Parallel chords of lengths 12 cm and 16 cm lie on opposite sides of the centre of a circle of radius 10 cm. The distance between them (in cm) is: 1

(a) 15 **(b)** 14 **(c)** 13 **(d)** 16

8. Parallel chords of lengths 12 cm and 16 cm lie on the same side of the centre of a circle of radius 10 cm. The distance between them (in cm) is: 1

(a) 3 **(b)** 2 **(c)** 4 **(d)** 1

9. **Assertion (A):** The opposite angles of a cyclic quadrilateral are supplementary. 1
Reason (R): The opposite angles of a cyclic quadrilateral are equal.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A chord of a circle can be longer than its diameter. 1
Reason (R): The diameter is the longest chord of a circle.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Two circles of radii 17 cm and 25 cm intersect, and the common chord is 30 cm. Find the distance between the centres if they lie on opposite sides of the chord. **2**

12. $ABCD$ is a cyclic quadrilateral and $\angle ABD = 27^\circ$, $\angle DBC = 49^\circ$. Find $\angle ABC$ and $\angle ADC$. **2**

13. Two parallel chords of lengths 12 cm and 16 cm lie on the opposite sides of the centre of a circle of radius 10 cm. Find the distance between them. **2**

14. Two circles of radii 25 cm and 29 cm intersect, and the common chord is 40 cm. Find the distance between the centres if they lie on opposite sides of the chord. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two circles of radii 25 cm and 26 cm intersect in a common chord of length 48 cm. Find the distance between their centres if they are on **3**
(i) opposite sides,
(ii) the same side of the common chord.

16. Parallel chords of lengths 14 cm and 48 cm in a circle of radius 25 cm. Find the distance between them in both possible positions. **3**

17. AB is a diameter of a circle; C, D lie on the circle on the same side of AB so that $ABCD$ is a cyclic quadrilateral with $\angle DAB = 52^\circ$. Find $\angle BCD$, $\angle ADB$ and $\angle ABD$. **3**

OR

Two circles of radii 25 cm and 30 cm intersect in a common chord of length 48 cm. Find the distance between their centres if they are on **3**
(i) opposite sides,
(ii) the same side of the common chord.

18. Parallel chords of lengths 30 cm and 40 cm in a circle of radius 25 cm. Find the distance between them in both possible positions. **3**

OR

AB is a diameter of a circle; C, D lie on the circle on the same side of AB so that $ABCD$ is a cyclic quadrilateral with $\angle DAB = 63^\circ$. Find $\angle BCD$, $\angle ADB$ and $\angle ABD$. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A circular track has centre O. Points A and B lie on the track such that the arc AB subtends $\angle AOB = 110^\circ$ at the centre. A point C lies on the major arc AB and a point E lies on the minor arc AB. **5**
(i) Find $\angle ACB$.
(ii) What is the angle subtended at any point of the circle by a diameter?
(iii) If D is another point on the major arc, find $\angle ADB$.
(iv) Find $\angle AEB$ and give the reason.

20. In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 20)^\circ$, $\angle B = (2y + 12)^\circ$, $\angle C = (4x + 10)^\circ$ and $\angle D = (4y + 6)^\circ$. Find x, y and all four angles. **5**

OR

A circle of radius 25 cm has a chord AB of length 14 cm. Find the distance of AB from the centre O, the area of $\triangle OAB$ and the perimeter of $\triangle OAB$. **5**

END OF QUESTION PAPER — Total: 40 Marks